Below are **Accenture-style interview answers** tailored for a **Machine Learning Engineer (Fresher/Entry-Level)**. I've also aligned them with your projects (**FoodSynthNet** and **Loan Defaulter Risk Prediction**) so you can use them directly in interviews.

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1. Tell me about yourself and why you want to be a Machine Learning Engineer at Accenture.

Answer

Good morning, and thank you for giving me this opportunity.

My name is **Shaik Afrin**, and I am pursuing my Bachelor of Technology in Computer Science Engineering at St. Ann's College of Engineering and Technology with a CGPA of **8.81**.

I have a strong interest in Artificial Intelligence and Machine Learning because I enjoy solving real-world problems using data.

During my internships, I worked on Python, Machine Learning, Data Science, and Full Stack Development, where I gained practical experience in data preprocessing, model development, and web application deployment.

One of my major projects is **FoodSynthNet**, a Generative AI-based food recognition and nutrition monitoring system. In this project, I used Conditional GANs to generate balanced food image datasets and developed a Flask web application that predicts food items and provides nutritional information.

Another project is **Loan Defaulter Risk Prediction**, where I built a Random Forest model to predict loan default risk and deployed it through a Flask application.

I want to join **Accenture** because it is a global leader in AI and digital transformation. The opportunity to work on large-scale AI solutions, collaborate with experienced professionals, and continuously learn emerging technologies aligns perfectly with my career goals. I believe my technical skills, project experience, and eagerness to learn will help me contribute effectively to Accenture's AI initiatives.

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2. Write a SQL query to find the top 5 customers by revenue in the last 30 days.

Answer

```sql
SELECT
    c.customer_id,
    c.customer_name,
    SUM(o.total_amount) AS total_revenue
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
WHERE o.order_date >= CURRENT_DATE - INTERVAL '30 days'
GROUP BY c.customer_id, c.customer_name
ORDER BY total_revenue DESC
LIMIT 5;
```

Explanation

- JOIN combines customers and orders.
- WHERE filters orders from the last 30 days.
- SUM() calculates total revenue.
- GROUP BY groups revenue per customer.
- ORDER BY DESC sorts highest revenue first.
- LIMIT 5 returns only the top five customers.

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3. How would you clean a dataset with 20% missing values in a key column?

Answer

The approach depends on the business problem and the importance of the missing data.

First, I would analyze why the values are missing—whether they are random or follow a pattern.

If the missing values are numerical, I would use:

- Mean imputation if the data is normally distributed.
- Median imputation if the data contains outliers.

For categorical columns, I would use:

- Mode imputation.

If the missing values represent meaningful information, I might create a new category such as "Unknown."

If the missing percentage is very high and the column is not important, I may remove the column.

If the affected rows are very few and cannot be recovered, I would remove those rows after evaluating the business impact.

Finally, I would validate that the preprocessing improves model performance.

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4. Describe a complex data analysis project you completed end-to-end.

Answer (STAR Method)

Situation

I developed **FoodSynthNet**, a Generative AI system for food recognition and nutrition monitoring.

Task

The challenge was that food image datasets were imbalanced, which reduced classification accuracy.

Action

I collected and preprocessed food images, generated synthetic images using Conditional GANs to balance the dataset, trained the model, evaluated its performance, and integrated it into a Flask web application that displays nutritional information and health recommendations.

Result

The project improved classification performance on underrepresented food classes and provided an end-to-end solution for food recognition and nutrition analysis. It also strengthened my skills in Generative AI, data preprocessing, and deployment.

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5. Explain the difference between LEFT JOIN and INNER JOIN.

Answer

INNER JOIN

Returns only matching records from both tables.

Example:

Customers

ID	Name
1	Alice
2	Bob

Orders

CustomerID	Order
1	Laptop

Result

Name	Order
Alice	Laptop

Bob is not included because he has no order.

LEFT JOIN

Returns all rows from the left table and matching rows from the right table.

Result

Name	Order
Alice	Laptop
Bob	NULL

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6. Write a Python function using Pandas to calculate a 7-day moving average.

Answer

```
import pandas as pd

def moving_average(df, column):
    df['7_day_avg'] = df[column].rolling(window=7, min_periods=1).mean()
    return df
```

Explanation

- rolling(window=7) creates a 7-day window.
- mean() calculates the average.
- min_periods=1 ensures the first few rows also receive values.

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7. Accenture has seen a 15% drop in a key metric this week. How do you diagnose the issue?

Answer

I would follow a structured approach:

1. Verify the accuracy of the data to rule out data pipeline issues.
2. Compare the metric with previous weeks.
3. Segment the data by region, customer type, device, or channel.
4. Check whether a recent deployment or product release affected the metric.

5. Look for external factors such as holidays or outages.
6. Analyze related KPIs to identify correlations.
7. Summarize findings and recommend corrective actions.

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8. Explain data normalization and when it matters.

Answer

Normalization scales numerical features to a common range, typically **0 to 1**.

Formula:

```
(X - Min) / (Max - Min)
```

It is important for algorithms that rely on distances, including:

- KNN
- K-Means
- Neural Networks
- Support Vector Machines

Normalization ensures that features with larger values do not dominate the model.

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9. How would you build a BI dashboard to track Accenture's key business metrics?

Answer

My approach would be:

1. Meet with stakeholders to understand business goals.
2. Define KPIs such as revenue, customer growth, project completion, and customer satisfaction.
3. Collect data from different sources.
4. Clean and transform the data using ETL processes.
5. Design a data model.
6. Build dashboards using Power BI or Tableau.
7. Add filters, drill-downs, and visualizations.
8. Schedule automatic refreshes and monitor performance.

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10. Where do you see data analytics heading in the next 3–5 years and how are you preparing?

Answer

I believe data analytics will become more AI-driven and real-time.

Some major trends include:

- AI-powered analytics
- Generative AI for business intelligence
- Real-time streaming analytics
- Cloud-native analytics platforms
- Data Mesh architecture
- Explainable AI
- Self-service analytics

To prepare, I am continuously improving my knowledge in Machine Learning, Generative AI, Python, SQL, cloud technologies, and working on hands-on AI projects. I also stay updated through online courses, technical blogs, and practical experimentation with modern AI tools.

Interview Tip

Keep each answer to **60–90 seconds** unless the interviewer asks for more detail. Structure your responses clearly, use examples from your projects, and explain both the technical approach and the business impact.



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